Central Banks and Financial Integration: Evidence from the Eurosystem Collateral Framework*

Pia Hüttl
DIW Berlin

Matthias Kaldorf
Deutsche Bundesbank

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Abstract

Can central banks policies contribute to deepen financial integration? In this paper, we exploit that the Eurosystem started to accept cross-border bank loans as collateral for its lending operations in January 2007. Banks holding a large share of newly eligible loans significantly increase their total credit supply. However, affected banks only supply more cross-border credit when participating in loan syndicates and not when acting as lead arrangers. Since harmonized collateral policy does not mitigate credit frictions between banks and firms, it improves financial integration only indirectly through intensified syndicate participation ("arm's length debt"), rather than promoting direct cross-border lending through "relationship lenders".

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*E-Mail: phuettl@diw.de (Hüttl) and matthias.kaldorf@bundesbank.de (Kaldorf, Corresponding Author). Contact Address: Wilhelm-Epstein-Str. 14, 60431 Frankfurt am Main, Germany.
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1 Introduction

Increasing financial market integration remains a major policy objective in Europe (Draghi, 2024). Due to the large degree of bank dependence in the euro area, one key objective of an European financial market union is to make firm funding conditions independent of the local banking sector's health. However, it remains an open question which policy instruments can contribute to deeper financial market integration in Europe (ECB Committee on Financial Integration, 2024) and what role central banks can play in this context (Altavilla et al., 2024).

In this paper, we study central bank collateral policy as a potential candidate to deepen financial market integration. In 2007, the Eurosystem harmonized its list of eligible collateral, making cross-border bank loans eligible for Eurosystem standing facilities. One stated goal of this policy was to "enhance the level playing field in the euro area, further promoting equal treatment for [...] issuers" of eligible loans (ECB Monthly Bulletin, 2006). Whether such a policy successfully increases financial market integration depends crucially on banks' credit supply decision, i.e. whether banks extended more and larger loans to firms located in other euro area countries. To test this empirically, we use loan-level data from the euro area syndicated loan market and exploit heterogeneity in banks' loan portfolios prior to the collateral framework shock.

We document that banks with a high exposure to cross-border euro area loans increase their loan supply by 8% after the framework change, compared to a control group of banks with low exposure. While this result is consistent with prior literature, our key novel result concerns the international lending dimension. We distinguish between loans where banks act as lead arranger in a syndicate, i.e. where they screen and monitor borrowers, and where they merely participate in a syndicate by supply funds but delegating screening and monitoring to the lead arranger (see Gustafson et al. (2021) and the references therein). We find that affected banks increase their cross-border lending when participating in loan syndicates, i.e., they supply more credit to previously ineligible firms. By contrast, affected banks only increase loan supply domestically when they act as lead arrangers, i.e., to firms that were already eligible prior to the single collateral list. This finding is in line with the presence of severe screening and monitoring frictions in *direct* cross-border lending, which the collateral eligibility of loans cannot overcome. Banks respond by intensifying *indirect* cross-border lending through syndicate participation, where such credit frictions are arguably less relevant. This suggests central bank collateral frameworks only partially contribute to deeper financial integration.

In January 2007, the ECB introduced a *single collateral list* specifying which assets euro

area banks can use as collateral in refinancing operations with the Eurosystem. After this collateral framework change, banks were able to pledge loans extended to borrowers in the whole euro area. Before 2007, each national central bank specified different collateral eligibility criteria, according to which banks could at most pledge domestic loans. To identify banks that are affected by the collateral framework shock, we use their issuance history of newly eligible loans. Specifically, we construct a bank-level measure using the loan issuance to newly eligible firms, i.e. *non-domestic euro area* firms, prior to the announcement of the single list in June 2005. Consistent with the literature, we perform a median split along the *Affected*-measure to classify banks into affected and unaffected banks and compare changes in the lending of affected banks relative to the control group of unaffected banks.[1]

We provide support for our identifying assumption that unaffected banks are a counterfactual for the lending of affected banks in the absence of the collateral framework shock. A possible confounding event is the proximity to the financial crisis unfolding after the burst of the US subprime bubble in the third quarter of 2007. While the early announcement of the single list in 2005 indicates that the collateral framework shock is a crisis-unrelated policy change, we still identify affected banks through their cross-border activity on the (euro area) syndicated loan market. Hence, affected banks might also be more active on the US market and would then change their credit supply in 2007 and 2008 for reasons unrelated to the collateral framework shock. To alleviate this concern, we keep the Post-event window very short in the baseline, even though the positive lending effects are quite persistent over time. Furthermore, we test for differential pre-trends, also employing the method proposed by Rambachan and Roth (2023) and, in addition, present two falsification tests. On one hand, we show that internationally active banks, located in the EU but outside the euro area, do not expand credit supply relative to their unaffected peers, controlling for loan demand. On the other hand, we use an ineligible asset class (revolving credit lines) as a placebo treatment indicator and find no credit supply effect by affected banks either.

We also enrich our baseline difference-in-difference specification along several dimensions. The treatment could lack random assignment, for example if large banks are more active internationally or less profitable banks could depend more on central bank facilities. Therefore, we include a large set of bank level controls and include bank country $\times$ time effect. Another concern is that affected and unaffected banks might face different lending opportunities after the collateral framework shock. To address this concern, we add firm $\times$ quarter fixed effects, which absorb loan demand, along the lines of Khwaja and Mian (2008). Once loan demand

[1] Our approach is related to the literature on the bank lending channel of monetary policy. This literature typically finds that bank liquidity conditions shape the credit supply response to a monetary policy shock, see Kashyap and Stein (2000), Gambacorta (2005) and Jiménez et al. (2014) among others. The collateral framework shock allows us to study the transmission of shocks to bank liquidity conditions themselves.

is controlled for, we find that affected banks increase their quarterly loan supply by 8% (corresponding to almost 36 million USD each quarter), which is in line with the credit supply effects of liquidity shocks originating on the interbank market (Iyer et al., 2013; Mitchener and Richardson, 2019). Furthermore, affected banks are more likely to extend new loans at the extensive margin.

The large importance of syndicated loans for cross-border lending in the euro area (Doerr and Schaz, 2021) prompts us to distinguish between affects banks' credit supply as lead arranger and as participant. Maintaining the distinction between participants and lead arrangers, we re-estimate our baseline specification for three sub-samples, based on the borrower location: domestic, non-domestic euro area, and non-euro area. As expected, we find no evidence of additional credit supply to non-euro area firms, which remained ineligible under the single list. Regarding firms in the euro area, we find stark differences between lead arrangers and participants. Affected banks increased credit supply to (newly eligible) firms in other euro area countries by around 8% when acting as participants of a new loan origination, but did not significantly change their domestic credit supply. By contrast, when acting as lead arrangers in a specific deal, affected banks experienced a significant but quantitatively small increase in cross-border lending by 0.9% and instead direct most of the additional credit supply to domestic borrowers (29.6%). These differences remain highly significant in a triple difference specification.

We show that the collateral eligibility of cross-border loans has merely stimulated cross-border lending *indirectly* through loan syndicate participation and not by strengthening borrower-lender relationships. To the best of our knowledge, the importance of distinguishing *direct* bank-to-firm relationships from syndicate participation is a novel finding in the literature on cross-border lending. The literature has suggested information asymmetries (Van Nieuwerburgh and Veldkamp, 2009), cultural differences (Giannetti and Yafeh, 2012), and country-specific bank regulation and corporate bankruptcy laws (Davydenko and Franks, 2008) as barriers to international loan market integration, and these frictions are arguably present in the euro area loan market. Hence, such barriers appear to severely dampen the transmission of collateral eligibility to direct bank-to-firm lending. By contrast, these frictions can reasonably assumed to be absent in the interaction between syndicate members, consistent with the much larger effect on indirect cross-border lending through intensified syndicate participation. In line with these considerations, we generally observe no increase in new bank-firm relationships at the extensive margin after the policy change, i.e. the entire credit supply effect is driven by repeated relationships.

The stark heterogeneity in international lending effects informs the ongoing discussion about the European banking and capital markets union, and the role central bank policies

can play in it.[2] As documented in Hoffmann et al. (2022), cross-border lending has been largely restricted to the interbank market, which is an indirect form of financial integration. Indeed, interbank market funding turned out to be a fragile financing source during the European debt crisis. To underscore the relevance of our loan level findings, we demonstrate that the *aggregate* credit supply drop during the financial crisis was much more pronounced for participating banks than for lead arrangers, in particular in the subset of cross-border loans. Lead arrangers are usually in charge of monitoring firms, resembling relationship lending, while credit supply through syndicate participation resembles "arms' length debt". Taken together, the single collateral list only partially improved financial integration, since it merely stimulated cross-border lending through relatively fragile syndicate participation and not by strengthening borrower-lender relationships.

Having discussed the limited cross-border effects of harmonized collateral policy, we study the role of bank liquidity constraints, bank risk-taking and real effects at the firm sector. The purpose of these additional tests is to show that the well-known transmission channels of (unconventional) monetary policy are also present for the single collateral list. This mitigates concerns that the limited cross-border effect is simply a consequence of a generally limited transmission of this specific shock. At the loan level, we find larger effects within the sub-sample of liquidity constrained banks. At the bank level, we show that banks increase loan supply to riskier borrowers as measured by the loan-volume weighted distance-to-default (Merton, 1974). An alternative interpretation of this finding is that banks extend loans to firms with tighter borrowing constraints (Farre-Mensa and Ljungqvist, 2015). As a complementary approach, we subset borrowers into firms active in tradable and non-tradable sectors along the lines of Müller and Verner (2023) and only find positive credit supply effects for the non-tradable sector. This is consistent with the risk-taking channel of collateral policy first documented by Van Bekkum et al. (2018). Different from prior literature, we document a risk-taking channel of unconventional monetary policy in normal times.[3]

[2]Both the 2008 financial crisis and the 2011 sovereign debt crisis revealed vulnerabilities of the euro area to asymmetric shocks, which can (partly) be attributed to the lack of financial market integration (Lane, 2012). The banking system's failure to absorb adverse shocks was at the heart of the 2008 and 2011 crises (Martin and Philippon, 2017). The 2008 financial crisis manifested itself in Europe through dry-ups for periphery banks funding themselves through the interbank market. In contrast, bank holdings of risky domestic sovereign debt were an important driver of the 2011 debt crisis. It is reasonable to assume that an integrated banking system would have softened the impact of either crisis on the real economy. Martinez et al. (2022) show that a financial market union that makes private sector funding conditions independent from the local banking system's health improves cross-country risk-sharing, see also the discussion in Constâncio (2014). More recent examples of such vulnerabilities are the asymmetric effects of the Covid19 pandemic and global supply chain disruptions on euro area sovereign bond markets, which prompted the European Commission to release a new Capital Market Union action plan in 2021.

[3]Related to this result, Jasova et al. (2021) document risk-taking effects of LOLR policies, Delis and Kouretas (2011) establish risk-taking effects of interest rate policy, Grosse-Rueschkamp et al. (2019) doc-

At the firm level, we demonstrate that the collateral framework shock did not merely induce a reallocation of loan financing from unaffected to affected banks, i.e. the policy change had real effects. We show that firms with a large exposure to affected banks increase their employment and tangible assets. Our results are consistent with findings in Paravasini (2008), Chodorow-Reich (2014), Amiti and Weinstein (2018), or Jasova et al. (2021) and are in line with standard theories of credit supply, (unconventional) monetary policy, and firm borrowing constraints. After an expansionary shock, banks' additional credit supply is directed to riskier and more credit-constrained borrowers, who use the additional credit supply to increase their real activity.

Finally, it is noteworthy that, different from essentially all other commonly used shocks to collateral frameworks in the empirical literature, the single list was introduced during a period of low financial stress. Thus, our paper establishes that the transmission channels of unconventional monetary policy through the banking sector are also at play during conventional times. This might bear important implications for future implementation of monetary policy in the euro area, as the ECB announced in its recently concluded operational framework review that it will continue to rely on "fixed-rate tenders with full allotment against broad collateral".[4]

Related Literature Our findings mainly contribute to two strands of research. First, we add to the literature studying the relationship between financial market integration and the real economy, which has been studied extensively using bank deregulation in the United States, see Morgan et al. (2004), Karakaya et al. (2022) or Goetz and Gozzi (2022) among others. In an euro area context, Buch (2003) shows that the European Union's Banking Directive did not overcome national fragmentation in cross-border lending, while Kalemli-Ozcan et al. (2010) document that the euro's effect on financial integration is mainly due to the elimination of currency risk and only to a small share by regulatory convergence. Both findings consistent with our results at the loan level. Burietz and Ureche-Rangau (2020) and Hoffmann et al. (2022) use country level data to establish a generally low level of *direct* cross-border lending and show that there was sizable indirect lending through the interbank market. Emter et al. (2018) document that cross-border lending fell sharply after the global financial crisis in 2008, in particular interbank loans. Bruche and Suarez (2010) propose a model of international lending via interbank markets that exhibits fragility with potentially large negative real implications. Therefore, it has been argued that the absence of a banking

ument elevated bank risk-taking after the Eurosystem's CSPP. Bruno and Shin (2015) stress the role of bank risk-taking for the international transmission of US monetary policy and Dinger and te Kaat (2020) demonstrate that international capital flows increase bank credit supply and risk taking.

[4]The full statement is available under this link.

and capital markets union was an important driver of the European debt crisis (see for example Lane, 2012 and the references therein). We contribute to this literature by showing that central bank collateral policy can affect cross-border lending via participation in loan syndicates, which is arguably less stable than a genuine borrower lender relationship but also less fragile than international bank-to-bank lending (Hoffmann et al., 2022). [5]

Second, by employing a shock to the collateral framework, our paper extends the literature on the effects of central bank operational frameworks.[6] Van Bekkum et al. (2018) document that changes in collateral eligibility concerning residential mortgage backed securities affects bank lending and risk-taking behavior in the mortgage market. Delatte et al. (2019) find that the Banque de France's Additional Credit Claims program has an positive impact on credit supply to French firms. Using a similar dataset, Harpedanne de Belleville (2023) documents positive effects on credit supply and employment. Mésonnier et al. (2021) document that such eligibility translates also into a relative reduction in rates for newly eligible bank loans. Cahn et al. (2024) use the ECB's very long term refinancing operations to study credit supply to single- and multi-bank firms. Pelizzon et al. (2024) also use properties of the ECB collateral framework in their empirical strategy. Using the collateral eligibility of corporate bonds, they find that eligibility reduces secondary market bond yields and increases bond issuance at the firm level. Our paper contributes to the literature by focusing on international lending effects of collateral policy.

The remainder of our paper is organized as follows. Section 2 describes the institutional setting, the collateral framework shock, and the dynamics of direct and indirect cross-border lending. We introduce our dataset in Section 3. Section 4 presents our empirical strategy and results. Section 5 concludes.

[5]The positive cross-border effect in response to the collateral framework shock is consistent with studying the role of bank liquidity for cross-border lending. Peek and Rosengren (2000) and Schnabl (2012) study international spillovers of adverse shocks to bank assets. Cetorelli and Goldberg (2011) study the transmission of bank shocks from developed to emerging economies using country-level data. Giannetti and Laeven (2012) demonstrate that banks disproportionately decrease international credit supply when funding conditions in their home market deteriorate. Popov and Udell (2012) provide evidence for downsides of financial integration by exploring how adverse liquidity shocks to international banks decrease SME lending in Eastern Europe through local subsidiaries. Giannetti and Jang (2024) argue that foreign lenders are more likely to extend credit supply prior to banking crises. In contrast, Doerr and Schaz (2021) find that international lenders maintain a higher credit supply during banking crises in borrower countries. Barbosa et al. (2018) stress the role of banking stress for international monetary policy spillovers.

[6]This literature is largely empirical. However, a small literature studies the role of collateral policy as central bank instrument through a theoretical lens, see for example Ashcraft et al. (2011), Koulischer and Struyven (2014), and Cassola and Koulischer (2019).

2 Institutional Setting

On 22 July 2005, the ECB announced the introduction of a *single collateral list*, applicable to the whole euro area, specifying which assets banks can pledge as collateral to obtain central bank funding through main refinancing operations or the marginal lending facility.[7] The single list came into effect on 1 January 2007. Previously, the collateral eligibility of bank assets was determined following a two-tier system. Tier-one assets consisted of marketable debt instruments, mostly government bonds, fulfilling euro area-wide eligibility criteria. The eligibility of tier-two assets was specified by national central banks, allowing them to incorporate idiosyncrasies of the respective domestic banking sector in the collateral framework. For a detailed review of the Eurosystem Collateral Framework, we refer to Nyborg (2017) and Bindseil et al. (2017), while Tabakis and Tamura (2013) discuss the collateral eligibility of credit claims, such as syndicated loans, in the Eurosystem.

Under the single list regime, the ECB directly specifies eligible assets for all euro area banks. Furthermore, the single list regime established the eligibility of bank loans extended to all borrowers located in the euro area, subject to a minimum requirement on loan size and firm rating. This also applied to syndicated bank loans, where different banks form a syndicate to jointly lend to a single borrower. The syndicate includes one lead bank and a number of participating banks. Lead arrangers are those members of a syndicate typically negotiate credit conditions, conduct due diligence, and monitor firms (Dennis and Mullineaux, 2000 and Ivashina and Scharfstein, 2010). Participants are usually not in direct contact with the borrower, but merely supply credit via the lead arranger. Importantly, valuation haircuts applied to pledged collateral do not differ between lead arrangers and participants, such that we can focus on loan eligibility throughout the analysis.

Collateral Policy and Financial Integration Harmonizing collateral policy drastically increased the eligibility of (cross-border) loans as collateral across the euro area banking sector. Prior to the single list, most national central banks (Germany, Austria, Spain, France, Ireland and the Netherlands) already accepted bank loans to domestic companies as collateral, while cross-border loans were not accepted by any national central bank. The additional eligible collateral, thus, largely stems from cross-border (syndicated) loans. Notably, the expansion of the set of eligible assets was not achieved by a relaxation of minimum rating requirements, which is in contrast with the ECB's expansionary policy measures undertaken later, such as a reduction of the minimum rating requirement from A to BBB during the financial crisis 2008 and a temporary suspension of minimum rating requirements during the

[7] The official announcement on 22 July 2005 contained details on the inclusion of bank loans, see ECB (2005).

euro area debt crisis and in April 2020. The ECB Monthly Bulletin (2006) explicitly states the following objectives of switching to a single list regime:

> *The aims of the single list are to* ***enhance the level playing field in the euro area, further promoting equal treatment for counterparties and issuers****, and to increase the overall transparency of the collateral framework. Moreover, the single list takes into account the fact that, with increasing collateralization in private wholesale markets and relatively high consumption of collateral by the Eurosystem, there are now* ***competing demands on the collateral holdings*** *of banks. More generally, by increasing the liquidity of an entire asset class, such as bank loans, the single list of collateral fosters the* ***smooth functioning of the euro area financial system.***
> ECB Monthly Bulletin, 2006, page 76.

The ECB Monthly Bulletin (2006) suggests that both the lack of financial market integration and the shortage of high-quality collateral were identified as major issues already prior to the financial crisis unfolding in 2008. First, reducing collateral scarcity was deemed necessary to ensure approximately frictionless trade among banks in the euro area financial system. Second, and more closely related to the real economy, the ECB intended to "enhance the level playing field [...] for issuers", i.e., promote cross-border lending.

Collateral Policy and Banks' Pledging Behavior As a prerequisite step, to achieve either objective, collateral policy needs to change banks' pledging behavior which, in turn, might also affect their loan supply. Sauerzopf (2007) provides suggestive evidence along these lines based on collateral pledged by Austrian banks. This is consistent with aggregate usage of non-marketable assets as collateral in ECB refinancing operations over time (Figure 1). Before 2007, both the absolute amount of non-marketable assets and their share of total collateral use was stagnant at around 35 bn EUR, or 4% of total collateral. By the end of 2007, the usage of non-marketable assets more than doubled to 109 bn EUR, and its share of total collateral increased to almost 11%.

While Figure 1 does not provide a bank-specific analysis of pledging behavior, an aggregate increase is indicative of banks' willingness to use non-marketable assets in Eurosystem operations. This bears several implications for bank behavior on the loan market. *First*, even if banks do not intend to pledge a specific loan, they might still prefer to hold eligible loans for precautionary reasons, for example to self-insure against adverse liquidity shocks. In technical terms, banks submit eligible assets into so-called collateral portfolios or collateral accounts, such that these assets can be mobilized quickly if they are needed as collateral.

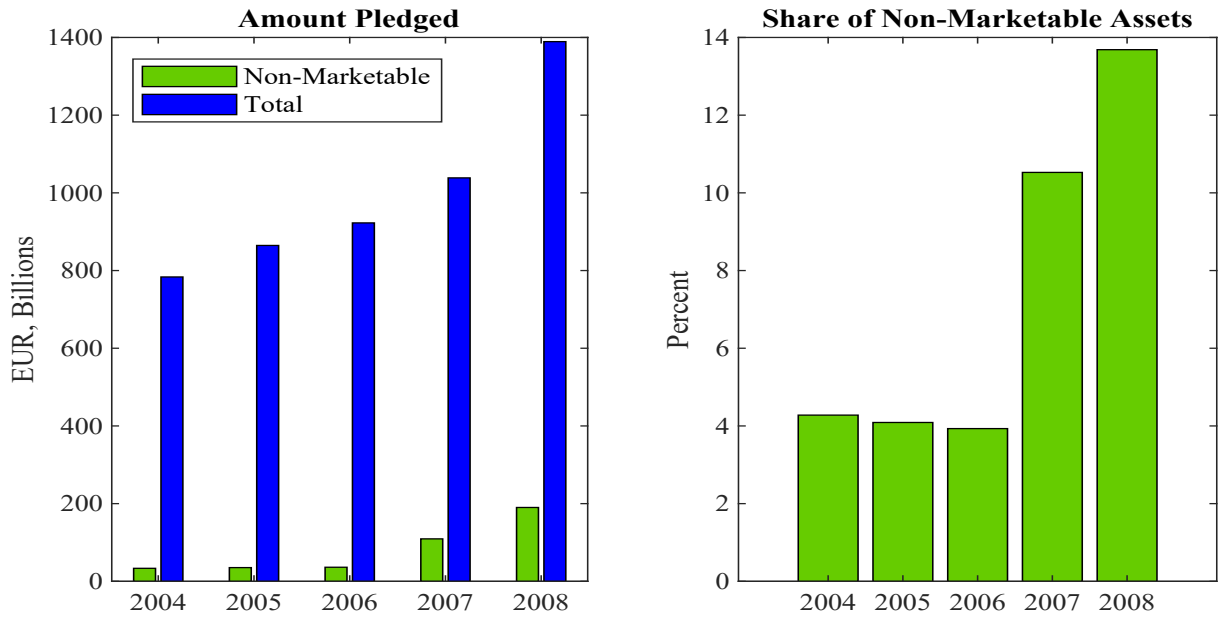

Figure 1: Non-Marketable as Collateral over Time. The left panel displays the evolution of aggregate collateral pledged with the ECB over time (blue) and aggregate non-marketable assets pledged with the ECB (green) in billions of EUR. The right panel displays non-marketable assets as a share of total collateral. Data source: European Central Bank.

Put differently, the eligibility into collateral portfolios matters for the pricing of an asset. Consistent with this idea, Barthélemy et al. (2018) show that, even though banks pledged more assets into collateral accounts than required *on aggregate*, around 10% of euro area banks hit their collateral constraints between 2011 and 2016. *Second*, the eligibility of non-marketable assets can have a positive spillover effect on bank holdings of high-quality liquid assets, such as government bonds. As argued in Pelizzon et al. (2024), there is effectively no private wholesale funding market for corporate sector assets in the euro area. If loans can be pledged at the Eurosystem, high-quality liquid assets can be used more effectively on the wholesale funding (repo) market instead.[8] In both cases, the collateral framework shock should have positive effects on banks' net worth. Our loan level analysis tests whether banks that were strongly affected by the shock take advantage of this positive effect and increase their credit supply.

Financial Integration and Syndicate Composition Before turning to the loan level analysis, we inspect the euro area syndicated loan market and its composition over time.

[8]Positive spillover effects from "freeing up" high-quality liquid assets are explicitly mentioned in ECB (2006), page 9. Choi et al. (2021) provide a theoretical analysis of this mechanism. Related "crowding-in" effects have been documented by Arce et al. (2020). They show that the ECB's corporate sector purchase programme enabled firms to switch from loan towards bond financing. This switch allowed banks to reallocate funds towards formerly constrained firms without bond market access, who were therefore not directly affected by the CSPP.

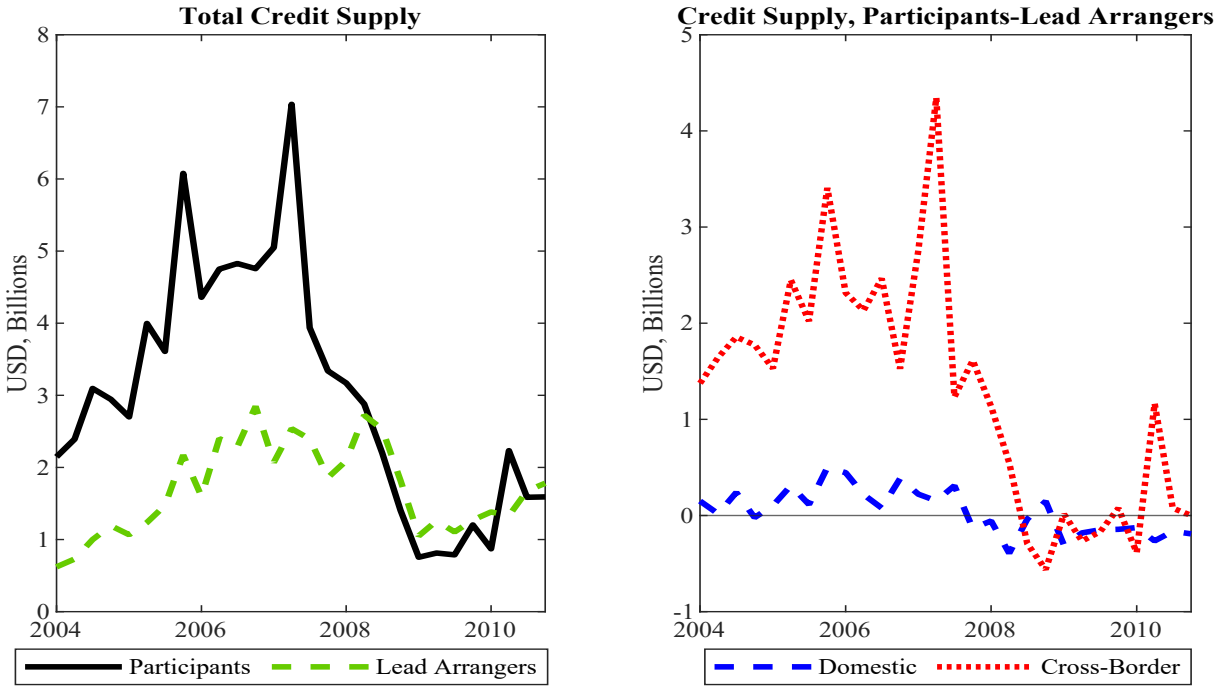

Figure 2: Credit Supply Decomposition over Time. The left panel shows the aggregate amount of new loans in each quarter for participants (solid black line) and lead arrangers (dashed green line) in billions of USD. The right panel shows the difference in aggregate loan issuance between participants and lead arrangers, for domestic loans (dashed blue line) and for cross-border loans (dotted red line).

Figure 2 plots total credit by lead arrangers (the "relationship lenders") and syndicate participants (providers of "arm's length debt") over time. The left panel suggests that total credit supply by participating banks collapsed by more than 80% during the financial crisis (solid black line), while lead arrangers provided much more stable funding to the real sector (dotted green line).

This retrenchment was particular pronounced for cross-border lending. The right panel of Figure 2 decomposes the credit supply differential between participants and lead arrangers - i.e. the difference between the black and green lines in the left panel - into domestic and cross-border lending. The dotted red line shows that the level of international credit supply by participants dropped slightly below the level of lead arrangers, after exceeding it by up to five billion USD between 2004 and 2007. By contrast, domestic credit supply by participants hardly declined during the crisis. In this sense, lending through syndicate participation exhibits a strikingly similar pattern to interbank lending and is arguably a less stable funding source for the real sector. Successful financial integration - one stated goal of the policy change - should hence promote *direct* cross-border lending, by forming or deepening borrower-lender relationships. Therefore, our empirical analysis explicitly distinguishes between credit supply by lead arrangers and participating banks.

3 Data

Our analysis is based on syndicated loan market data from DealScan, where we observe the borrowing firm and all participating banks at the loan level. We focus on syndicated loans for three reasons. First, when the single collateral list was introduced, the collateral eligibility was subject to a minimum requirement on the loan size, such that SME loans were typically ineligible. Second, syndicated loans make up 75% of bank-to-firm cross-border lending in the euro area (Doerr and Schaz, 2021). Third, the euro area credit registry was established well after the introduction of the single list. We complement loan level information with bank data from CapitalIQ and firm data from Compustat.

Loan Data As a first step, we restrict the sample to non-financial firms and to commercial, savings, cooperative and investment banks. We decompose syndicated loan deals into loan portions provided by each lender to obtain loan level data. Whenever DealScan provides information on lending shares of each bank, we use this information to split loan volume accordingly. In other cases, we closely follow Schwert (2018) to estimate lending shares, who proposes to use hedonic regressions to back out loan shares. Specifically, we regress loan shares on the loan size, the number of participants, borrower and lender sales in a Tobit model and then predict loan shares for transactions where they are missing. Transactions with deal status 'canceled', 'suspended', or 'rumor' are removed and all loan nominations transformed into million USD using the spot exchange rate at origination, provided by DealScan. If after this allocation procedure the loan portion is smaller than 10,000 USD, we drop the observation to remove erroneously small loans. Importantly, we do not need to observe credit line drawdowns, since credit lines are generally ineligible as collateral with the Eurosystem and we use them in a Placebo test.

We then aggregate all loan issuances between a bank-firm combination to obtain bank i's loan issuance to firm j in quarter t, which we define as a bank-firm-quarter observation. Total loan volume in a given quarter is the sum of all new loans issued by bank i to firm j. In doing so, we only account for syndicated loan issuances, disregarding the redemption profile. Table 1 presents summary statistics on the bank-firm-quarter level over the sample period Q1 2006 to Q4 2007. The average loan issuance from bank i to firm j in quarter t amounts to 451.13 million, the average spread over LIBOR to 204 basis points, and the average maturity of the loans to around 7 years. All loans in our sample have at least one designated lead arranger. We also observe whether a loan is secured. The share of loans issued to domestic and other euro area firms amounts to 54%. Of these, approximately 60% are issued to domestic firms, while the remaining 40% go to firms headquartered in other

euro area countries. 46% of all loans are extended to non-euro area firms. Domestic firms are defined as firms which have their headquarters in the same country as the corresponding bank.

Bank Characteristics To control for bank characteristics, we match the banks included in the DealScan dataset with bank balance sheet data from CapitalIQ. Panel A of Table 2 presents summary statistics for all euro-area banks in the period prior to the framework shock (Q1 2006 - Q4 2006) included in our sample. On average, banks hold 59% loans and 26% securities over total assets. On the funding side, deposits make up 41% and equity 5% of total assets on average. The Return on Equity amounts to 15.3% on average across the sample period. Panel B of Table 2 presents evidence on the difference in bank characteristics between affected and unaffected banks using univariate t-tests. Affected banks are banks which have an above median share of euro area loan issuances in their syndicated loan issuances in the period prior to the announcement (Q1 2003 - Q2 2005). Affected banks are similar in terms of size, cash, deposit, and equity ratios, but have a slightly higher return-on-equity (13.7% vs 17.3%), significant at the 10% level.

Firm Variables We obtain annual firm balance sheet data for European firms from Compustat. We aggregate the quarterly loan data from DealScan to the firm-year level and match borrowers in DealScan with firms in Compustat. The matching is based on Chava and Roberts (2008), updated in April 2018. Combining those two databases reduces observations, since not all firms have balance sheet data available con Compustat, especially not smaller ones. Eventually, we obtain a sample of 1,121 firms. Variables are winsorized at the 1st and 99th percentile. As customary, financials (SIC codes 6000-6999) are dropped. Panel A of Table 3 shows summary statistics for the full sample while Panel B establishes that more and less exposed firms did not differ significantly along the dimensions we use as a control or outcome variable in 2006, the year prior to the single list's implementation.

4 Empirical Strategy and Results

We use a standard difference-in-differences set-up to compare the lending of banks affected by the collateral framework change to the lending of unaffected banks. Section 4.1 describes how we identify banks that are most affected by the shock using the share of loans that were issued *before* the framework change was announced and became eligible after the single list was implemented. Our baseline result in Section 4.2 establishes that banks with a large share of newly eligible loans on their balance sheet increase their credit supply relative to banks

with a smaller share of such assets. We provide two falsification tests and several robustness checks in Section 4.3 to corroborate our main findings. We then examine the cross-border lending effects in Section 4.4 before turning to the extensive margin in Section 4.5. Lastly, we show that the positive shock to the liquidity of bank assets affects risk-taking at the bank level and real outcomes at the firm level in Section 4.6 and Section 4.7, respectively.

4.1 Identifying Affected Banks

Banks are classified into *affected* and *unaffected* based on their loan issuance histories to newly eligible borrowers prior to the single list's announcement.[9] Assuming that the collateral framework shock has larger effects on banks which were already actively issuing loans that became eligible under the single list, we identify affected banks according to their issuance history from Q1 2003 until Q2 2005, the last quarter prior to the announcement. To accurately measure bank-level exposure at the group level, the subsidiary structure of each group i has to be taken into account, since subsidiaries directly pledge assets at the NCB in the country where they are chartered. We denote the set of all subsidiaries of bank group i by $\mathcal{K}^i$ and we restrict our attention to euro area subsidiaries.

We cumulate subsidiary k's issuance of newly eligible loans over the period prior to the single list's announcement. Let c_j denote the home country of firm j and let $\mathcal{C}_k^{new}$ the set of countries where loans became newly eligible under the single list regime. For a subsidiary located in Germany, the set of newly eligible countries is given by $\mathcal{C}_k^{new} = \mathcal{C}^{ea} \backslash \{DE\}$, since loans to German firms were already eligible under the NCB-regime in Germany. By contrast, for a subsidiary located in Italy, we simply have $\mathcal{C}_k^{new} = \mathcal{C}^{ea}$. The subsidiary-level *Affected*-measure is then given by

$$\text{Affected}_k \equiv \frac{\sum_{t \in \mathcal{T}} \sum_{j \in \mathcal{J}^{ea}} \mathbf{1}\{c_j \in \mathcal{C}_k^{new}\} \cdot L_{kjt}}{\sum_{t \in \mathcal{T}} \sum_{j \in \mathcal{J}^{ea}} L_{kjt}} , \tag{1}$$

where L_{kjt} are loans issued by subsidiary bank k to firm j in quarter t. The set of pre-announcement dates is denoted by $\mathcal{T}$ = [Q1 2003, Q2 2005], while $\mathcal{J}^{EA}$ is the set of all euro area firms in our sample.

As we are interested in financial integration across euro area countries, we mostly focus lending from ultimate owners in one country to firms in another country, we then aggregate the subsidiary-level *Affected*-measure to the group level. For each subsidiary, the *Affected*-

[9] For banks headquartered in Austria, France, Germany, Ireland, the Netherlands and Spain, all loans to firms in other euro area countries are *newly eligible*. For banks in Belgium, Greece, Italy and Portugal, all euro area loans, including loans to domestic firms, are *newly eligible*.

measure is weighted by the subsidiary k's share of the group-level loan portfolio:

$$\text{Affected}_i \equiv \frac{\sum_{t\in\mathcal{T}}\sum_{j\in\mathcal{J}^{ea}} L_{kjt}}{\sum_{k\in\mathcal{K}^i}\sum_{t\in\mathcal{T}}\sum_{j\in\mathcal{J}^{ea}} L_{kjt}} \cdot \text{Affected}_k \ . \tag{2}$$

The *Affected*-measures at group and subsidiary level have a correlation of around 90%. Internationally active parent banks tend to have internationally active subsidiaries as well.

We perform a sample split of banks along the *Affected*-measure (2) and interpret all banks with an above-median share of newly eligible euro area loans in their portfolio as affected. This identification strategy is common in the literature, see for example Huber (2018), Van Bekkum et al. (2018) or Grosse-Rueschkamp et al. (2019). The results are robust to using total assets instead of total loans in the denominator, see Appendix B.5. While we mostly consider results at the bank group level, we show in Appendix B.3 that our results also hold when estimating the credit supply effect at the subsidiary level, irrespective of using the subsidiary level or the aggregated group level *Affected*-measure. Interestingly, estimating subsidiary-level loan supply using the group level *Affected*-measure yields a larger credit supply effect. This points to the existence of bank group-internal capital markets (De Haas and Van Lelyveld, 2010) and also provides further justification to using the group level *Affected*-measure, which appears to be the more important determinant of credit supply.

4.2 Baseline Specification

Formally, we estimate the effect of the single list's inclusion on credit supply at the bank-firm-quarter level using the following specification:

$$ln(L_{ijt}) = \beta_1 \text{Affected}_i(0/1) \times \text{Post07}_t + \beta \text{X}_{i,t-4} + \mu_{ij} + \nu_{jt} + c_{it} + c_{jt} + \epsilon_{ijt} \ . \tag{3}$$

Here, $Post07_t$ is a dummy that indicates the single list regime. $X_{i,t-4}$ is a vector of loan and bank level controls, which are lagged by 4 quarters. μ_{ij} denotes bank $\times$ firm, ν_{jt} denotes firm $\times$ quarter, c_{it} denotes bank country $\times$ quarter, and c_{jt} denotes firm country $\times$ quarter fixed effects. The coefficient β_1 measures how affected banks respond to the collateral framework shock relative to unaffected banks. Standard errors are clustered at the bank level, the level at which the treatment occurs, to adjust for serial correlation within treated units.

Our identifying assumption is that banks less active in cross-border euro area loan syndication provide a valid counterfactual for bank behavior in the absence of a framework shock. For a causal interpretation of the estimated effect and the coefficient size, several concerns need to be dismissed. *First*, the heterogeneous lending behavior of affected and unaffected banks could be caused by other factors than collateral eligibility of their loan portfolio, for

example bank size or funding conditions. To address this concern, we include the log of assets, return on equity, as well as equity, cash, securities and deposits ratios over total assets, all lagged by four quarters. These control variables are standard in the empirical literature on the bank-lending channel (Kashyap and Stein, 2000). In Panel B of Table 2, we show that affected and unaffected banks are largely similar along these dimensions.

Second, the treatment could lack random assignment if it were based on a variable that has a differential impact on affected and unaffected banks and correlates with the collateral framework shock, such as a large exposure to international credit markets. We address this by designing two placebo tests using always ineligible assets, described in Section 4.3. Furthermore, in Appendix B.6, we also demonstrate that our results are robust to excluding German banks, which are traditionally very active on international credit markets.

We also include bank country × quarter fixed effects to ensure that our results are not driven by heterogeneity in the treatment of bank loans under the NCB-regime. Since the NCBs in Belgium, Greece, Italy and Portugal did not accept domestic bank loans before the policy change, banks headquartered in these countries benefited relatively more from the single collateral list. While bank country × time fixed effects ensure a fair comparison with banks headquartered in countries where domestic loans were already accepted before the single list, it should be noted that banks headquartered in Belgium, Greece, Italy, and Portugal only make up 11% of all loans in our sample, see Table A.2.

Third, the pool of potential borrowers might differ for affected and unaffected banks, and hence banks would face different lending opportunities after the shock. To address this concern, we include bank-firm fixed effects and firm-quarter fixed effects. The former captures relationship lending from the same bank to the same firm. The latter allows identification of loan supply, as we compare the lending of affected and unaffected banks to the same borrower, absorbing loan demand (Khwaja and Mian, 2008). In an alternative specification, we replace firm × quarter fixed effects by country × industry × quarter fixed effects, following Degryse et al. (2019).

Table 4 presents the baseline result from estimating equation (3), where we apply some conservatism in truncating the sample in 2007 Q4 to exclude the Great Financial Crisis and additional collateral framework changes in response to the crisis as potential confounders. However, the lending effects are persistently positive until 2009 Q4, as we show in Figure 3. The first three columns show the results from increasing the stringency of fixed effects. In column (1), we only use bank × firm fixed effects to compare lending of affected and unaffected banks to the same firm j before and after the collateral framework change in January 2007. Both the coefficient of interest (0.085) and the $Post07_t$ indicator (0.179) are positive. Important potential confounders that are not captured in this simplistic specification are

country-specific credit demand and supply factors (Acharya et al., 2018) if affected banks are not equally distributed across euro area countries. In column (2), we control for time-varying country differences by including bank country × quarter and firm country × quarter fixed effects. In addition, we add firm × quarter fixed effects to control for loan demand (Khwaja and Mian, 2008) and relationship fixed effects. We find a positive treatment effect (6.1%), which is significant at the 1% level.

Column (3) shows a specification with loan and bank level controls that refine the comparison between treatment and control group. Banks with a higher securities ratio supply fewer loans on average, which is however not robustly significant across specifications. Other bank level controls are insignificant across specifications. Additional loan supply is significantly larger for secured loans, other loan level controls are at not significant. Including controls slightly increases the size of β_1 to 8%. To put this effect into perspective, we multiply the coefficient by the (full sample) average loan volume from bank i to firm j at time t (see Table 1). Affected banks increase their loan supply by $0.08 \cdot 451 = 36$ mn USD each quarter. In column (4), we replace firm × quarter fixed effects by country × industry × quarter fixed effects (Degryse et al., 2019), which barely changes the results. The coefficient remains highly significant and is slightly smaller (7.3%). In all specifications, the magnitude of the estimated credit supply effect is similar to Van Bekkum et al. (2018).

So far, we have considered the sample with all loan observations. In column (5), we only include interactions between a given bank-firm pair if bank i has been a lead arranger in at least one loan extended to firm j, while column (6) includes interactions where bank i has never been a lead arranger in loans extended to firm j. We focus on the most stringent specifications and find that the effects are substantially larger for lead arrangers (10.6%) than for participants (3.8%).

To test the parallel trend assumption, we first plot coefficient estimates of the baseline specification for loan supply from Q3 2005 to Q4 2009. Figure 3 shows that there are no significant pre-trends between affected and unaffected banks, as coefficient estimates are statistically insignificant and hover around zero. After the collateral framework change, the lending activity of affected banks becomes positive and significant relative to the control group of unaffected banks. As long as confounding factors affect both types of banks in the same way, for example an accommodating monetary policy stance in the early 2000s, they are canceled out by the difference-in-differences approach. Second, we build on Rambachan and Roth (2023) to check the robustness of our treatment effect. Figure 4 shows that the treatment effect appears to be robust to moderate deviations from the parallel trends assumption ($\hat{M}$ is significant up to 0.4). In other words, we can rule out a null effect unless we allow for violations of parallel trends 0.4 times larger than the largest violation in the

pre-period.

4.3 Falsification Tests

Our classification of affected banks is based on their cross-border lending history prior to the collateral framework shock. While treatment and control group do not significantly differ in terms of observable characteristics (Table 2), affected banks might still have an unobserved larger lending capacity than their unaffected peers. Furthermore, affected banks might also be more active on the international loan market in general and the US market in particular. In the context of the financial crisis unfolding in 2008, affected banks might therefore exhibit a differential credit supply reaction that is unrelated to the collateral framework shock.[10] While we generally use a relatively short window around the January 2007 event, from January 2006 to December 2007 in our baseline, we additionally conduct two *falsification tests* exploiting that the Eurosystem single list is relevant (i) only for banks headquartered in the euro area and (ii) only applies to fixed-term loans.

For the first test, we build a placebo group of affected banks residing outside the euro area, but inside the EU, which are mainly banks headquartered in the UK. Similar to equation (1), we compute the share of euro area loans over total loans for every subsidiary of a non-euro area headquartered bank group and aggregate the subsidiary-level *Affected*-measure using the subsidiary share in the group loan portfolio. As before, we use a median split along the *Affected*-measure to test whether the collateral framework shock has a differential impact on foreign lenders that would be heavily affected by the single list's introduction, relative to their unaffected peers.

For the second falsification test, we make use of the fact that only term loans are pledgeable as collateral under the single list. Reconstructing the *Affected*-measure at subsidiary and bank group level using the issuance history of revolving credit lines thus provides us with another placebo *Affected*-measure that we use to re-estimate equation (3). Using an ineligible asset class to conduct a falsification test is in line with Van Bekkum et al. (2018). Table 5 shows that, for both falsification tests, the coefficients on the $Affected_i(0/1) \times Post07_t$ interaction term are insignificant or even negative across all specifications. This suggests that the credit supply effects in our baseline specification indeed reflect the collateral framework shock and are not driven by other factors that affect treatment and control group in heterogeneous ways. Additionally, we run several robustness test to further corroborate our main

[10] Puri et al. (2011) show that European banks exposed to U.S. sub-prime market exhibit a more pronounced decrease of credit supply. In contrast to this narrative, we expect the collateral framework shock to stimulate loan issuance of affected banks. Therefore, a larger exposure to international loan markets would bias the results downwards.

findings. Appendix B.1 shows that affected banks also lowered the interest rate spreads after the collateral framework change, while Appendix B.2 illustrates that there were no significant announcement effects.

4.4 International Lending Effects

Building on our baseline specification, we now examine the international lending effects of the collateral framework change. Specifically, we estimate our baseline specification (3) for different sub-samples based on the borrower location vis-a-vis banks. A positive coefficient β_1 for the sub-samples of other euro area borrowers would indicate an increase in cross-border lending. Due to the large importance of the syndicated loan market for cross-border lending in the euro area, we explicitly distinguish between the credit supply of affected banks when they act as lead arrangers and as syndicate participant. Importantly, we construct these sub-samples deal by deal, which does not change the classification of banks as affected or unaffacted. For the lead arranger sub-sample, a positive coefficient on the $Affected_i(0/1) \times Post07_t$ interaction term in the international sub-sample would indicate a strong direct cross-border lending effect. For the participant sub-sample, a positive coefficient suggests that international lending effects are not associated with direct bank-to-firm relationships and rather resemble indirect cross-border lending, as the borrower-lender relationship is intermediated by the lead arranger. To control for loan demand in the three sub-samples, we use country × industry × time fixed effects (Degryse et al., 2019), which enables us to estimate our baseline specification on smaller sub-samples.

Panel A of Table 6 shows that the effect on direct cross-border lending is positive and significant, but much smaller (0.9%) compared to the full sample, see column (2). Instead, the single list's effect is an order of magnitude larger for lending to firms headquartered in the affected lead banks' home country (29.6%), see column (1). For participating banks, the opposite picture emerges. Credit supply to domestic firms is not significantly affected, while loan issuance to firms in other euro area countries increases by around 8%, see columns (4) and (5). These coefficients are significantly different from each other, as we show in a triple difference specification using the borrower location as additional dummy (Table B.4).

In column (3) and (6), we display the additional credit supply of affected banks to firms located outside the euro area when they act as lead arrangers and participants, respectively. This is a helpful benchmark for international lending by affected and unaffected banks, as loans to non euro area firms are generally not affected by Eurosystem collateral policy. Consequently, we find no effect for lead arrangers or participants. In unreported results, we find that the additional domestic lending by lead arrangers is significant in a triple-difference

specification using loans to non euro area firms as second layer of control, i.e. by comparing the coefficients in columns (1) and (3). Likewise, the additional lending to other euro area firms for participants is significantly larger than the lending of participants to non euro area firms, i.e. by comparing the coefficients in columns (5) and (6).

Importantly, lead arrangers are generally *more* responsive to the single list's introduction, as the sample split in the full sample has shown (columns (5) and (6) of Table 4). This makes the absence of economically meaningful international lending effects for lead arrangers even more noteworthy. The small international lending effect is consistent with the idea that collateral eligibility does not alleviate frictions in direct bank-to-firm relationships, such as asymmetric information or country-specific bankruptcy laws. Thus, affected lead arrangers predominately increase their credit supply domestically, where such frictions are less relevant compared to lending internationally. At the same time, the positive shock to the liquidity of bank assets induces affected banks to increase international lending as participants, as the shock still makes it more attractive to extend such loans. However, the additional cross-border lending is not associated with intensifying direct borrower-lender relationships. Instead, this happens indirectly through syndicate participation.

Reassuringly, estimating the same sample split on always ineligible credit lines as yet another falsification test also yields no significant effects. Panel B of Table 6 shows that there is no credit supply effect for (always ineligible) credit lines for lead arrangers or participants. This suggests that the collateral framework shock was indeed driving the international lending effects and that we are not picking up a general retrenchment of cross-border lending by internationally active banks at the onset of the Great Financial Crisis.

4.5 Extensive Margin

Up to this point, our analysis focused on the intensive margin of credit supply. In this section, we extend our analysis to the extensive margin and build a balanced panel by adding zeros to all bank-firm-quarter triples when there was no loan issuance. On this panel, we estimate a linear probability model, where the dependent variable is an indicator variable equal to one if bank i supplied a loan to firm j in quarter t, irrespective of the loan size. Since Eurosystem collateral eligibility is also subject to a minimum loan size, this analysis alleviates the concern that affected banks issue larger loans, but do so less often after the framework change. Column (1) of Table 7 shows that affected banks have a 0.2 percentage point larger probability of extending a loan after the single list was introduced. Combined with a positive intensive margin, a positive effect in the extensive margin specification points towards an increase in total credit supply, since banks interact with firms more often and

supply more credit when they interact. In unreported results, we obtained similar results hold for the total margin, i.e. when estimating the baseline specification equation (3) over a balanced panel by adding zeros to all bank-firm-quarter triples when there was no loan issuance.

The extensive margin can also shed light on the effect of the collateral framework expansion to the formation of new bank-firm relationships. Such new relationships would also be indicative of a higher degree of financial integration in the largely bank-based euro area economy. We split the sample in new and repeated lending relationships. Comparing the results in columns (2) and (3) of Table 7 reveals that the extensive margin effects are entirely driven by lending in repeated relationships. The probability of extending a loan in a repeated relationship is 1.4 percentage point larger for affected banks after the treatment. By contrast, there is no significant effect on new relationships. When further splitting the sample between lead arrangers (columns 4 and 5) and participants (columns 6 and 7), we find positive effects in both sub-samples, although they are not significant for lead arrangers, where the sample size becomes quite small. Taken together, this suggest that the collateral eligibility of (cross-border) loans does not induce new bank-firm relationships. This is consistent with the limited international lending effects and jointly suggests that central bank collateral policy is unable to overcome real barriers to financial integration such as information asymmetries.

4.6 Bank Risk-Taking

The limited cross-border lending effects documented in the previous section could be a consequence of a generally limited transmission of this specific shock. To alleviate this concern, we investigate how the single list affects lending by banks that arguably benefit more from this shock, whether it increases bank risk-taking, and how it transmits to the real economy.

As a first step, we estimate our baseline specification on a sub-sample of more and less liquidity constrained banks. In the spirit of Kashyap and Stein (2000), we define banks with a below-median ratio of securities over total assets as more liquidity constrained. The left panel of Table 8 reveals that affected banks in this sub-sample increase their loan supply by around 9%, while firms with a higher securities ratio only expand lending by around 6%. The discrepancy is even stronger when using country $\times$ industry $\times$ time fixed effects. This is not surprising, as such banks benefit particularly from a relaxation in collateral policy and hence access to central bank lending facilities.

In a second step, we provide two distinct approaches to show that in particular high-risk and credit constrained firms are subject to funding inflows. First, we estimate equation (3) on two sub-samples split according to firms being classified as tradable or non-tradable. Firms

active in the non-tradable sector are typically associated with higher credit risk and tighter borrowing constraints (Müller and Verner, 2023). We classify firms based on primary SIC codes reported in DealScan (Giannetti and Jang, 2024).[11] The coefficient on $Affected_i(0/1) \times \text{Post07}_t$ is positive and significant for non-tradables but insignificant for tradables, see Panel A of Table 9. This result is consistent with Berg et al. (2022), who show that the positive liquidity shock to the banking system induced by the ECB's corporate bond purchase programme had positive credit supply effects almost exclusively in the real estate sector, i.e. an important part of the non-tradable sector.

We also take a complimentary approach at the bank level and use the distance-to-default (Merton, 1974) of banks' loan portfolios after the shock as dependent variable. The advantage of this measure is that it can be easily computed from Compustat balance sheet and stock price data. For details on the distance-to-default, its theoretical foundations and practical implementation, we refer to Gilchrist and Zakrajšek (2012), Bharath and Shumway (2008) and the references therein. Specifically, we estimate effect of the single list regime on the distance-to-default of bank i's loan portfolio at quarter t:

$$D2D_{it} = \gamma_1 \text{Affected}_i(0/1) \times \text{Post07}_t + \gamma \text{X}_{i,t-4} + c_{jt} + \epsilon_{ijt} \ . \tag{4}$$

Since our analysis is carried out at the bank level, we define country-quarter fixed effects based on the bank headquarter in (4). The results are shown in Panel B of Table 9. The coefficient γ_1 is significantly negative, i.e. the distance-to-default declines after the shock for affected banks and the riskiness of the loan portfolio increases. This result is consistent with risk-taking effects reported in Van Bekkum et al. (2018), who show that banks were investing into riskier residential mortgages following a relaxation in eligibility requirements for residential mortgage backed securities. More generally, our results are consistent with the risk-taking channel of monetary policy (Jiménez et al., 2014). Interpreting the distance-to-default as a measure of borrowing constraints (Farre-Mensa and Ljungqvist, 2015), we can also conclude that the real effects of central bank collateral policy are particularly pronounced for borrowing constrained firms. Different from most prior literature, our results document a risk-taking channel of unconventional monetary policy during normal times.

[11] The tradable sector mainly consists of manufacturing firms (SIC code 2000-3999), while the non-tradable sector includes construction (SIC code 1500-1799), wholesale and retail services (SIC code 5000-5999) and accommodation (SIC code 7000-7099).

4.7 Firm Level Results

Lastly, we study the real effects of the collateral framework change at the firm level using annual data. To identify firms affected by the collateral framework expansion, we construct a firm level *exposure* measure based on the share of loans obtained from affected banks between January 2005 and December 2006. Formally, the exposure measure is given by

$$\text{Exposed}_j = \frac{\sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}} \mathbf{1}\{i \in \mathcal{I}^{Affected}\} L_{ijt}}{\sum_{i\in\mathcal{I}}\sum_{t\in\mathcal{T}} L_{ijt}} , \tag{5}$$

where $\mathcal{I}^{Affected}$ is the set of affected banks according to the *Affected*-measure (2) and $\mathcal{T}$ = [Q1 2005, Q4 2006] is the pre-implementation sub-sample. We then perform a sample split of all firms along the *Exposed*-measure (5) and define all firms with an above-median as exposed.

To test whether the collateral framework shock translates into real outcomes at the firm level, we set up the following regression equation at the firm-year level:

$$y_{jt} = \delta_1 \text{Exposed}_j \times \text{Post}_{07,t} + \delta \text{X}_{j,t-1} + \mu_j + \chi_{jt} + \epsilon_{jt} \tag{6}$$

where y_{jt} are real sector outcomes at the firm-level, such as *ln(PPE)*, *ln(Employment)* and *ln(Sales)*. The relative change in property, plants, and equipment (PPE) is a standard proxy for tangible investment. Since tangible investment is typically financed using long-term debt, we expect the most pronounced effect at this margin. $X_{j,t-1}$ is a vector of firm level controls, which consists of ln(total assets), leverage (total debt over total assets), and return on assets (ROA), all lagged by one year. μ_j are firm and χ_{jt} industry $\times$ year fixed effects. Robust standard errors are clustered at the industry level, as there are enough clusters (90 industries) to reliably obtain a cluster-robust variance estimator. A positive coefficient δ_1 indicates positive real effects. Firms that are highly exposed to the collateral framework shock use the proceeds of higher loan take-up to increase their investment and employment and sales.

In Table 10, we show how the shock to liquidity of bank assets transmits into positive real effects. Interacting the firm level exposure dummy with the Post07$_t$-indicator variable reveals a positive and significant effect on tangible assets (around 10 percent) and employment (around 4 percent), once firm and industry $\times$ year fixed effects are included. While both effects are robust to including firm controls such as size, leverage, and return on assets, there are more pronounced effects on investment. In unreported results, we do not find significant effects on firm's debt maturity profile or their liquid asset holdings. Notably, the increase in real activity is measured in 2007 and 2008, which already contains the first quarters of the

financial crisis. This suggests that the real effects of the collateral framework shock were not short-lived or restricted to periods of overall benign credit conditions.

5 Conclusion

Since the euro's inception in 1999, improving euro area financial market integration has been a major policy objective, as it has played an important role in amplifying the 2008 Financial Crisis and the 2011 European debt crisis. However, it remains unclear which instruments effectively contribute to financial market integration. This paper focuses on one potential instrument to increase direct cross-border lending. In 2007, the ECB introduced an unified list of collateral, whereby the Eurosystem started accepting all loans granted to firms in the euro area as collateral. This single list of collateral was intended to create "a level playing field" for all issuers of eligible collateral across the euro area. In bank-dependent economies like the euro area, this aims at making firm access to credit independent of their local banking sector's health.

We find that the single list had a positive effect on credit supply by affected banks of around 8%, which is a magnitude that is consistent with the literature. Our key novel result is that this credit supply expansion is *not* accompanied by a generally positive effect on cross-border lending. We document positive international lending effects for participation in loan syndicates, while we do not find cross-border lending effects for direct lending from lead arrangers to firms. We also could not find any effects on new bank-firm relationships at the extensive margin. The lack of an effect on direct cross border lending is consistent with theories on real barriers to cross-border lending, such as country-specific insolvency laws or information frictions due to the lack of geographical proximity. These frictions can reasonably be assumed to be absent between syndicate participants and lead arrangers, which can explain the positive effect of international syndicate participation. Notably, collateral policy does not directly reduce contracting frictions between firms and banks, such that its contribution to a financial market union is small, at least until such real barriers are lifted. Policies explicitly targeted at reducing such frictions, for example harmonized corporate bankruptcy legislation, appear to be more promising in improving euro area financial integration.

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6 Tables and Figures

Table 1: Summary Statistics: Bank-Loan-Quarter Level. This table presents summary statistics on the bank-loan-quarter level. The sample period is Q1 2006 to Q4 2007. *Loan amount* is the loan issuance granted by euro-area lead arranger i to firm j at quarter t. The *spread* is calculated as basis points over LIBOR including annual fees. *Loan Deal Purpose* is a categorical variable with five categories: Other Corporate (1), Working Capital (2), Fixed Investment (3), Debt Repayment (4), M & A (5).

	mean	sd	min	max	count
Loan amount (mn USD)	475.73	743.51	8.96	3,176.25	10,117
Spread (bps)	209.28	120.51	25.00	550.00	10,117
Maturity (months)	82.38	36.36	3.00	515.00	10,021
Share of lead arrangers (%)	0.32	0.47	0.00	1.00	10,117
Loan Deal Purpose $\in 0, 5$	3.45	1.67	1.00	5.00	10,117
Loan Refinancing (0/1)	0.31	0.46	0.00	1.00	10,117
Loan Secured (0/1)	0.97	0.18	0.00	1.00	7,457
Distance to Default (%)	12.35	12.15	4.08	42.56	723
Foreign loans (%)	0.49	0.50	0.00	1.00	10,117
Euro area loans (%)	0.51	0.50	0.00	1.00	10,117
Of which					
Other euro area loans (%)	0.49	0.50	0.00	1.00	5,125
Domestic loans (%)	0.51	0.50	0.00	1.00	5,125

Table 2: Summary Statistics: Bank Level. Panel A presents summary statistics for all euro area banks included in the baseline sample from 2006, the year before the framework change. *Loans ratio* refers to gross loans outstanding and is defined - as all other ratios - over total assets and indicated in percent. The *Affected*-measure is defined according to equation (2). Panel B shows univariate t-tests from 2006, the year before the framework change, between less affected and more affected banks, based on a median split along the *Affected*-measure (2). ***, **, * denote significance at the 1, 5, and 10% level, respectively and critical values are adjusted for multiple testing using a Bonferroni correction.

Panel A: Full Sample

	Mean	SD	Min	Max	Count
ln(Total Assets)	11.43	1.49	8.50	14.17	56
Loans ratio	62.42	15.65	16.44	83.66	56
Securities ratio	23.38	17.96	5.99	81.25	56
Cash ratio	1.48	2.16	0.00	10.76	56
Deposit ratio	42.15	19.30	3.71	71.40	56
Equity ratio	5.52	2.74	2.18	16.40	56
Return on Equity (%)	14.34	4.87	4.85	20.91	56
Affected (%)	64.01	39.58	0.00	100.00	56
Affected (0/1)	0.52	0.50	0.00	1.00	56

Panel B: Multiple t-tests

	Less Affected		More Affected			
	Mean	N	Mean	N	Diff.	t-stat.
ln(Total Assets)	11.41	27	11.45	29	-0.043	-0.106
Loans ratio	64.58	27	60.60	29	3.973	0.931
Securities ratio	31.20	27	17.28	29	13.920	2.709**
Cash ratio	1.25	27	1.69	29	-0.439	-0.757
Deposit ratio	38.79	27	45.44	29	-6.654	-1.288
Equity ratio	4.70	27	6.27	29	-1.565	-2.214
Return on Equity (%)	12.65	27	15.91	29	-3.253	-2.626*

Table 3: Summary Statistics: Firm Level. This table presents summary statistics for the firm cross-section in 2005-2006, the years prior to the collateral framework change. *Loan* refers to the sum of syndicated loans issued to firm j in year t. *Leverage* is based on firm j's long term debt, *Liquidity* refers to operating net cash flows (Compustat item "ibc"). Both ratios are defined with respect to total assets and indicated in percentage points. The last two columns in Panel B shows the univariate t-statistic for a test of equal means between both groups in 2006, the year prior to the collateral framework shock. ***, **, * denote significance at the 1, 5, and 10% level, respectively and critical values are adjusted for multiple testing using a Bonferroni correction.

Panel A: Full Sample					
	Mean	SD	Min	Max	Count
ln(Loan amount)	2.86	3.47	0.00	12.28	1,121
ln(Total assets)	8.25	1.72	4.40	14.89	1,068
Leverage ratio	37.67	17.87	0.06	79.40	1,070
Liquidity ratio	3.15	6.77	-25.13	19.44	1,074
ln(Tangible assets)	6.86	2.22	1.59	13.51	1,042
ln(No of employees)	2.16	1.23	0.04	5.03	938
Exposed(0/1)	0.50	0.50	0.00	1.00	1,121
Exposed(%)	46.94	36.76	0.00	100.00	1,121

Panel B: Multiple t-tests						
	Less Exposed		More Exposed		Diff	t-stat
ln(Loan amount)	2.94	180	3.07	189	-0.132	-0.363
ln(Total assets)	8.30	170	8.13	180	0.170	0.931
Leverage ratio	37.42	172	37.87	178	-0.454	-0.237
Liquidity ratio	2.09	172	3.23	180	-1.142	-1.497
ln(Tangible assets)	6.97	169	6.69	172	0.276	1.160
ln(No of employees)	2.34	150	2.05	151	0.285	2.054

Table 4: Loan Supply: Loan-Quarter Level. This table presents the effect of the single list's introduction on credit supply. *Loan* is the loan issuance from bank i to firm j at quarter t. *Affected$_i$(0/1)* is based on a median split of banks along the *Affected*-measure, see equation (2). *Post07$_t$* indicates the single-list regime after January 2007. The sample split in Column (5) and (6) is between banks in the role of *lead* and *participant* when syndicating a loan. The control variables on bank and loan level are lagged by 4 quarters. Country-fixed effects are based on respective headquarters. ***, **, * denote significance at the 1, 5, and 10% level, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
					Lead	Participant
VARIABLES	ln(Loan Vol)	ln(Loan Vol)	ln(Loan Vol)	ln(Loan Vol)	ln(Loan Vol)	ln(Loan Vol)
Affected$_i$(0/1) × Post07$_t$	0.085	0.061***	0.080***	0.073***	0.106***	0.038**
	(0.073)	(0.019)	(0.022)	(0.017)	(0.035)	(0.019)
ln(Total Assets)			0.029	0.015	0.033	-0.062*
			(0.030)	(0.041)	(0.044)	(0.035)
Equity ratio			-0.015*	-0.031***	-0.010	0.005
			(0.009)	(0.011)	(0.011)	(0.006)
Return on Equity			-0.000	-0.000	-0.000	-0.002**
			(0.000)	(0.001)	(0.000)	(0.001)
Cash ratio			0.012**	0.015**	-0.006**	0.016
			(0.005)	(0.006)	(0.003)	(0.014)
Securities ratio			0.002***	0.001	-0.001	0.001**
			(0.001)	(0.001)	(0.001)	(0.001)
Deposit ratio			-0.000	-0.000	0.001	-0.001***
			(0.000)	(0.001)	(0.001)	(0.000)
Loan refinancing			0.452	0.413	0.670*	0.126
			(0.295)	(0.282)	(0.375)	(0.532)
Loan secured			0.418***	0.428***	0.468***	0.381***
			(0.089)	(0.091)	(0.121)	(0.141)
Loan purpose			0.143	0.120	0.098	0.165
			(0.093)	(0.076)	(0.114)	(0.131)
Post07$_t$	0.179***					
	(0.055)					
Observations	10,117	10,117	10,117	9,997	3,209	6,861
R-squared	0.851	0.862	0.863	0.864	0.866	0.860
Bank-level Controls	No	No	Yes	Yes	Yes	Yes
Loan-level Controls	No	No	Yes	Yes	Yes	Yes
Bank × Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
FirmCountry × Time FE	No	Yes	Yes	Yes	Yes	Yes
BankCountry × Time FE	No	Yes	Yes	Yes	Yes	Yes
FirmCountry × Industry × Time FE	No	No	No	Yes	No	No
Firm × Time FE	No	Yes	Yes	No	Yes	Yes
Cluster	Bank	Bank	Bank	Bank	Bank	Bank

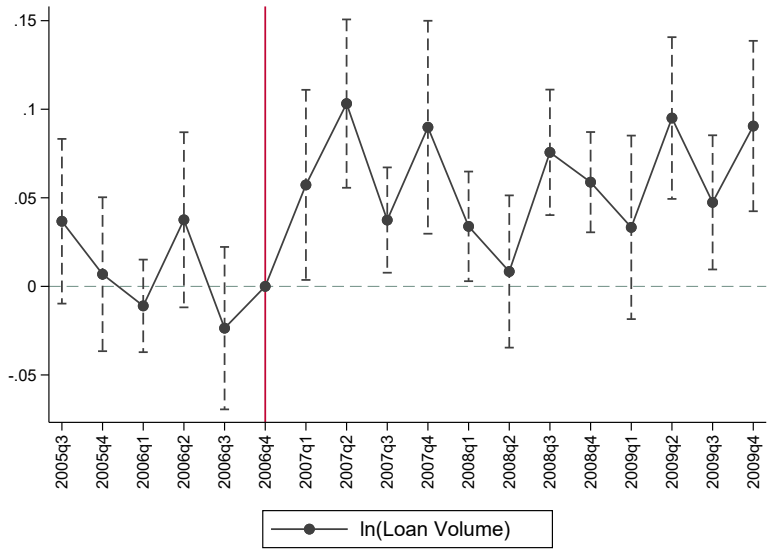

Figure 3: Parallel Trend Assumption. The figure is based on the following equation:

$$\log(L_{ijt}) = \sum_{\tau \neq Q4\ 2006} \beta_\tau \text{Affected}_i(0/1) \times \mathbf{1}\{\tau = t\} + \varepsilon_{ijt} \ ,$$

$\mathbf{1}\{\tau = t\}$ is a dummy variable that equals one in quarter t and 0 otherwise. Q4 2006, the quarter before the collateral framework shock, is excluded to estimate the dynamic effect. The dashed lines represent 90% confidence intervals, adjusted for bank level clustering.

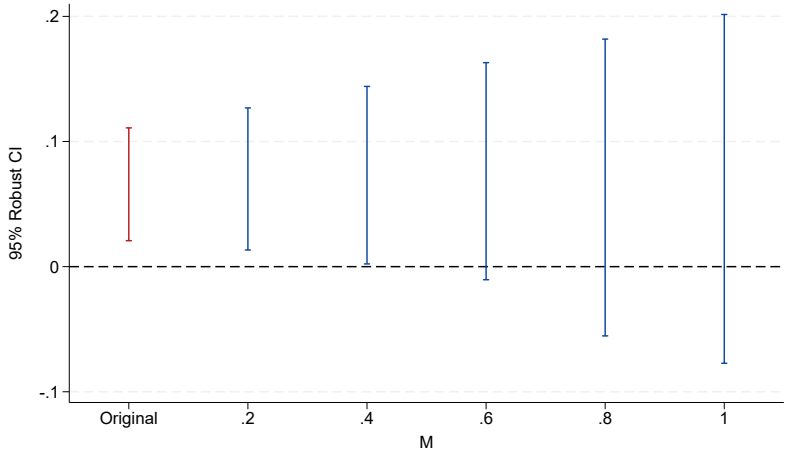

Figure 4: Sensitivity Analysis of Treatment Effect. The figure is based on Rambachan and Roth (2023). Instead of requiring that the parallel trend assumptions holds exactly, it introduces a *relative magnitude restriction*, which allows for the violation of parallel trends between 2006q4 and 2007q1 to be no more than $\hat{M}$ times larger than the worst pre-treatment violation of parallel trends. The figure shows robust confidence sets for the treatment effect in 2007q1 for $\Delta^{RM}(\hat{M})$ using different values of $\hat{M}$.

Table 5: Falsification Tests. Panel A presents the effect of the single list's introduction on credit supply of European banks outside the euro area, not affected by Eurosystem collateral policy. Panel B presents results for always ineligible credit lines. The analysis is based on data on the bank-loan-quarter level from Q1 2006 to Q4 2007. *Loan* is the loan issuance from bank i to firm j at quarter t. *Affected$_i$(0/1)* is based on a median split of banks along the respective placebo *Affected*-measure, see Section 4.3. *Post07$_t$* indicates the single-list regime after January 2007. The control variables on bank and loan level are lagged by 4 quarters. In Panel A, country-fixed effects are based on firm headquarters. ***, **, * denote significance at the 1, 5, and 10% level, respectively.

Panel A: Non-Euro Area Banks				
VARIABLES	(1) ln(Loan Vol)	(2) ln(Loan Vol)	(3) ln(Loan Vol)	(4) ln(Loan Vol)
Affected$_i$(0/1) × Post07$_t$	0.070	-0.079***	-0.016	0.009
	(0.076)	(0.023)	(0.088)	(0.074)
Observations	3,492	3,492	3,492	3,434
R-squared	0.857	0.860	0.861	0.862
Bank-level Controls	No	No	Yes	Yes
Loan-level Controls	No	No	Yes	Yes
Bank × Firm FE	Yes	Yes	Yes	Yes
FirmCountry × Time FE	No	Yes	Yes	Yes
FirmCountry × Industry × Time FE	No	No	No	Yes
Firm × Time FE	No	Yes	Yes	No
Cluster	Bank	Bank	Bank	Bank

Panel B: Credit Lines				
VARIABLES	(1) ln(Loan Vol)	(2) ln(Loan Vol)	(3) ln(Loan Vol)	(4) ln(Loan Vol)
Affected$_i$(0/1) × Post07$_t$	-0.026	-0.003	0.000	0.005
	(0.041)	(0.007)	(0.010)	(0.025)
Post07$_t$	0.148***			
	(0.026)			
Observations	6,505	6,052	6,052	6,291
R-squared	0.888	0.901	0.902	0.904
Bank-level Controls	No	No	Yes	Yes
Loan-level Controls	No	No	Yes	Yes
Bank × Firm FE	Yes	Yes	Yes	Yes
FirmCountry × Time FE	No	Yes	Yes	Yes
BankCountry × Time FE	No	Yes	Yes	Yes
FirmCountry × Industry × Time FE	No	No	No	Yes
Firm × Time FE	No	Yes	Yes	No
Cluster	Bank	Bank	Bank	Bank

Table 6: Cross-Border Loan Supply. This table provides sample splits according to borrower location vis-a-vis banks. The analysis is based on data on the bank-loan-quarter level from Q1 2006 to Q4 2007. *Loan* is the loan issuance from bank i to firm j at quarter t when it is a lead arranger (columns 1-3) or a participant (columns 4-6). *Affected*$_i$*(0/1)* is based on a median split of banks along the *Affected*-measure, see equation (2). *Post07*$_t$ indicates the single-list regime after January 2007. The control variables on bank and loan level are lagged by 4 quarters. Country-fixed effects are based on firm headquarters. ***, **, * denote significance at the 1, 5, and 10% level, respectively.

Panel A: Baseline

	Lead Arrangers			Participants		
	(1)	(2)	(3)	(4)	(5)	(6)
	Domestic	Other EA	Non EA	Domestic	Other EA	Non EA
VARIABLES	ln(Loan)	ln(Loan)	ln(Loan)	ln(Loan)	ln(Loan)	ln(Loan)
Affected$_i$(0/1) × Post07$_t$	0.296***	0.009*	0.018	-0.017	0.079***	-0.029
	(0.030)	(0.005)	(0.018)	(0.013)	(0.014)	(0.045)
Observations	943	800	1,445	1,224	2,134	3,459
R-squared	0.893	0.867	0.839	0.894	0.885	0.817
Bank-level Controls	Yes	Yes	Yes	Yes	Yes	Yes
Loan-level Controls	Yes	Yes	Yes	Yes	Yes	Yes
DomAff Control	Yes	Yes	Yes	Yes	Yes	Yes
Bank×Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Country×Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Country×Ind×Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Cluster	Bank	Bank	Bank	Bank	Bank	Bank

Panel B: Credit Lines

	Lead Arrangers			Participants		
	(1)	(2)	(3)	(4)	(5)	(6)
	Domestic	Other EA	Non EA	Domestic	Other EA	Non EA
VARIABLES	ln(Loan)	ln(Loan)	ln(Loan)	ln(Loan)	ln(Loan)	ln(Loan)
Affected$_i$(0/1) × Post07$_t$	0.031	0.025	-0.057	-0.000	-0.006	-0.016
	(0.031)	(0.022)	(0.090)	(0.000)	(0.006)	(0.022)
Observations	568	608	654	637	1,061	2,589
R-squared	0.933	0.936	0.854	0.870	0.912	0.871
Bank-level Controls	Yes	Yes	Yes	Yes	Yes	Yes
Loan-level Controls	Yes	Yes	Yes	Yes	Yes	Yes
DomAff Control	Yes	Yes	Yes	Yes	Yes	Yes
Bank×Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Country×Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Country×Ind×Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Cluster	Bank	Bank	Bank	Bank	Bank	Bank

Table 7: Extensive Margin: Repeated Vs New Bank-Firm Relationships. This table presents the effect of the single list's introduction on credit supply. The sample is extended into a balanced panel of firm-bank pairs at the quarterly frequency, including zeros where there was no interaction in the firm-bank pair and by replacing bank-firm fixed effects with bank fixed effects. The dependent variable *Pr(Loan)* equals one if bank i extended a loan to firm j at quarter t. $Affected_i(0/1)$ is based on a median split of banks along the *Affected*-measure, see equation (2). $Post07_t$ indicates the single-list regime after January 2007. *New* refers to firm-bank pairs interacting only once across the sample, *Repeated* to firm-bank pairs interacting more than once. ***, **, * denote significance at the 1, 5, and 10% level, respectively.

	Full Sample			Lead Arrangers		Participants	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	All	New	Repeated	New	Repeated	New	Repeated
VARIABLES	Pr(Loan)	Pr(Loan)	Pr(Loan)	Pr(Loan)	Pr(Loan)	Pr(Loan)	Pr(Loan)
$\text{Affected}_i(0/1) \times \text{Post07}_t$	0.002*	-0.000	0.014***	0.003	0.034	-0.003	0.012**
	(0.002)	(0.003)	(0.001)	(0.002)	(0.020)	(0.002)	(0.006)
Observations	137,600	112,032	20,480	27,016	2,616	81,864	13,184
R-squared	0.856	0.924	0.856	0.917	0.770	0.911	0.854
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country×Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm×Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster	Bank	Bank	Bank	Bank	Bank	Bank	Bank

Table 8: Constrained and Unconstrained Banks. This table presents the effect of the single list's introduction on credit supply of constrained and unconstrained banks. The sample split is along the median of a banks' securities to total asset ratio. *Loan* is the loan issuance from bank i to firm j at quarter t. *Affected*$_i$*(0/1)* is based on a median split of banks along the *Affected*-measure, see equation (2). *Post07*$_t$ indicates the single-list regime after January 2007. The control variables on bank and loan level are lagged by 4 quarters. Country-fixed effects are based on respective headquarters. ***, **, * denote significance at the 1, 5, and 10% level, respectively.

	Low Securities Ratio		High Securities Ratio	
	(1)	(2)	(3)	(4)
VARIABLES	ln(Loan Vol)	ln(Loan Vol)	ln(Loan Vol)	ln(Loan Vol)
Affected$_i$(0/1) × Post07$_t$	0.094**	0.185**	0.058***	0.076**
	(0.042)	(0.073)	(0.017)	(0.034)
Observations	4,972	4,955	5,062	5,044
R-squared	0.869	0.871	0.854	0.857
Bank-level Controls	Yes	Yes	Yes	Yes
Loan-level Controls	Yes	Yes	Yes	Yes
Bank × Firm FE	Yes	Yes	Yes	Yes
FirmCountry × Time FE	Yes	Yes	Yes	Yes
BankCountry × Time FE	Yes	Yes	Yes	Yes
FirmCountry × Industry × Time FE	No	Yes	No	Yes
Firm × Time FE	Yes	No	Yes	No
Cluster	Bank	Bank	Bank	Bank

Table 9: Risk-Taking Effects. Panel A provides a sample split according to borrower sector. The analysis is based on data on the bank-loan-quarter level from Q1 2006 to Q4 2007. *Loan* is the loan issuance from bank i to firm j at quarter t. *Tradable* are firms active in tradable industries (SIC code 2000-3999). *Non-tradable* are firms active in non-tradable industries (SIC code 5000-5999,6500-6599, 7000-7099). Panel B is based on bank-quarter level and uses the distance-to-default of bank i's loan issuance in quarter t as proxy for risk-taking. *Affected$_i$(0/1)* is based on a median split of banks along the *Affected*-measure, see equation (2). *Post07$_t$* indicates the single-list regime after January 2007. The control variables on bank and loan level are lagged by 4 quarters. Country-fixed effects are based on respective headquarters. ***, **, * denote significance at the 1, 5, and 10% level, respectively.

Panel A: Borrower Sector

	Tradable		Non Tradable	
	(1)	(2)	(3)	(4)
VARIABLES	ln(Loan)	ln(Loan)	ln(Loan)	ln(Loan)
Affected$_i$(0/1) × Post07$_t$	0.017	0.017	0.080***	0.034
	(0.011)	(0.011)	(0.022)	(0.143)
Observations	3,781	3,798	1,585	1,601
R-squared	0.879	0.879	0.774	0.775
Bank-level Controls	Yes	Yes	Yes	Yes
Loan-level Controls	Yes	Yes	Yes	Yes
Bank × Firm FE	Yes	Yes	Yes	Yes
FirmCountry × Time FE	Yes	Yes	Yes	Yes
BankCountry × Time FE	Yes	Yes	Yes	Yes
FirmCountry × Industry × Time FE	No	Yes	No	Yes
Firm × Time FE	Yes	No	Yes	No
Cluster	Bank	Bank	Bank	Bank

Panel B: Bank Risk Taking

	(1)	(2)
VARIABLES	D2D	D2D
Affected$_i$(0/1) × Post07$_t$	-0.016*	-0.007**
	(0.008)	(0.002)
Observations	245	245
R-squared	0.993	0.998
Bank-level Controls	No	Yes
Bank-Country × Time FE	Yes	Yes
Cluster	Bank	Bank

Table 10: Firm Level: Real Effects. This table provides results at the firm-year level from 2005 to 2008. The treatment variable *Exposed$_j$(0/1)* is based on a median split of firms along the *Exposed*-measure, see equation (5). *Post07$_t$* indicates the single-list regime after January 2007. The control variables on firm level are lagged by 1 year. ***, **, *, denote significance at the 1, 5, and 10% level, respectively.

VARIABLES	(1) ln(PPE)	(2) ln(Employ)	(3) ln(Sales)
Exposed$_j$(0/1) $\times$Post07$_t$	0.082*	0.052**	0.062*
	(0.047)	(0.022)	(0.034)
Observations	1,972	1,625	1,970
R-squared	0.980	0.988	0.980
Firm Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes
Cluster	Firm	Firm	Firm

A Additional Descriptive Statistics

Table A.1: Variable Definitions

Variable	Source	Description
Loan volume	DealScan	Loan volume to firm j by bank i in quarter t
Spread	DealScan	Spread over LIBOR including annual fees in basis points
Maturity	DealScan	Maturity of syndicated loan in months
Lead arranger	DealScan	Indicator variable equal to one if a bank acts as lead arranger
Ln(Loan)	DealScan	Logarithm of (one plus) the loan issuance from bank i to firm j at quarter t
ln(Total Assets)	CapitalIQ	Logarithm of one plus total assets
Loans ratio	CapitalIQ	Share of gross loans over total loans
Equity ratio	CapitalIQ	Share of equity over total assets (leverage ratio)
ROE (%)	CapitalIQ	Return on equity
Cash ratio	CapitalIQ	Share of cash and equivalents over total assets
Securities ratio	CapitalIQ	Share of investment securities over total assets
Deposit ratio	CapitalIQ	Share of deposits over total assets
Tradable	Compustat	Indicator equals one if firm j is active in tradable industries (SIC codes 2000-3999)
Non-tradable	Compustat	Indicator equals one if firm j is active in non-tradable industries (SIC codes 1500-1799, 5000-5999, 7000-7099)
Pr(Loan)	DealScan	Indicator variable that equals one if firm j obtains a bank loan in period t, and zero otherwise
ln(Total assets)	Compustat	Natural logarithm of one plus total assets
Leverage	Compustat	Ratio of long term debt to total assets
Liquidity	Compustat	Ratio of cash equivalents over total assets
Employment	Compustat	Number of employees, in thousands
ROA (%)	Compustat	Return on assets
PPE	Compustat	Plants, Property and Equipment
D2D	Compustat	Distance-to-default, computed over a one-year horizon

Table A.2: Summary Statistics: Distribution over Countries. This table splits loan-level observations over the full sample by bank headquarter country.

Country	Number of Loans	Frequency (%)
Austria	157	1.55
Belgium	276	2.73
France	3,002	29.67
Germany	2,788	27.56
Greece	36	0.36
Ireland	715	7.07
Italy	1,131	11.18
Netherlands	1,077	10.65
Portugal	81	0.80
Spain	854	8.44

B Additional Results

This section presents several additional empirical results, complementing our baseline specification. Specifically, we vary the outcome variable by considering credit spreads (Section B.1). We show that there are no significant announcement effects (Section B.2). Furthermore, we use the *Affected*-measure at the subsidiary level (Section B.3) and provide further results on the differences between lead arrangers and participants. Lastly, we show that our credit supply results hold when changing the definition of the *Affected*-measure (Section B.5) and when excluding German banks (Section B.6).

B.1 Loan Spreads

While we focus on the quantity dimension of credit supply - loan volume - in the main text,it is also possible to use the price dimension as a measure of credit supply, i.e. loan spreads. Therefore, we use the same specification to test the effect of the single list regime on loan spreads:

$$S_{ijt} = \beta_1 \text{Affected}_i(0/1) \times \text{Post07}_t + \gamma \text{X}_{i,t-4} + \mu_{ij} + \nu_{jt} + c_{jt} + \epsilon_{ijt} \ . \tag{7}$$

Since we expect affected banks to increase their credit supply, the coefficient β_1 should be negative when using spreads as dependent variable.

Second, Table B.1 shows that the response of loan spreads is consistent with our baseline results. Under the most stringed specifications in columns (3) to (5), the coefficient on $Affected_i(0/1) \times Post07_t$ is significantly negative at the 1%-level once loan demand is controlled for. The effect size of 16 basis points is comparable to Pelizzon et al. (2024) or Mésonnier et al. (2021) and is quite large given the full sample average spread of 200 basis points. Bank and loan level controls are mostly insignificant, only the loan purpose has a significantly positive effect on loan spreads.

Table B.1: Loan Spreads. This table presents the effect of the single list's introduction on loan spreads. *Affected$_i$(0/1)* is based on a median split of banks along the *Affected*-measure, see equation (2). *Post07$_t$* indicates the single-list regime after January 2007. The control variables on bank and loan level are lagged by 4 quarters. The sample split in Column (5) and (6) is between banks in the role of *lead* and *participant* when syndicating a loan. Country-fixed effects are based on respective headquarters. ***, **, * denote significance at the 1, 5, and 10% level, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
					Lead	Participant
VARIABLES	Spread	Spread	Spread	Spread	Spread	Spread
Affected$_i$(0/1) × Post07$_t$	-5.717	-8.161**	-6.814*	-7.395*	-11.971***	-2.915
	(10.738)	(3.568)	(3.751)	(3.783)	(3.292)	(4.084)
ln(Total Assets)			4.198	3.127	-11.570	35.967**
			(5.487)	(5.079)	(9.558)	(15.038)
Equity ratio			0.632	-0.195	0.491	-0.233
			(1.914)	(1.727)	(2.310)	(1.652)
Return on Equity			-0.163	-0.188	0.100	-0.216
			(0.154)	(0.148)	(0.112)	(0.138)
Cash ratio			0.028	0.154	4.056	-3.989**
			(0.698)	(0.743)	(2.400)	(1.884)
Liquidity ratio			0.404	0.325	0.603	0.248**
			(0.256)	(0.248)	(0.680)	(0.120)
Deposit ratio			-0.017	-0.018	-0.154	0.115
			(0.079)	(0.083)	(0.186)	(0.174)
Loan refinancing			-32.879	-35.986	-26.476	-52.258
			(27.146)	(25.105)	(25.218)	(41.730)
Loan secured			-18.008*	-18.746*	-21.620	-14.892**
			(9.414)	(9.424)	(15.267)	(6.405)
Loan purpose			3.922	2.407	17.291**	-10.149
			(4.465)	(3.919)	(7.795)	(13.796)
Post07$_t$	-7.842*					
	(4.632)					
Observations	10,117	10,117	10,117	9,997	3,209	6,861
R-squared	0.749	0.761	0.761	0.760	0.782	0.746
Bank-level Controls	No	No	Yes	Yes	Yes	Yes
Loan-level Controls	No	No	Yes	Yes	Yes	Yes
Bank × Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
FirmCountry × Time FE	No	Yes	Yes	Yes	Yes	Yes
BankCountry × Time FE	No	Yes	Yes	Yes	Yes	Yes
FirmCountry × Industry × Time FE	No	No	No	Yes	No	No
Firm × Time FE	No	Yes	Yes	No	Yes	Yes
Cluster	Bank	Bank	Bank	Bank	Bank	Bank

B.2 Announcement Effects

The single list and the future eligibility of cross-border bank loans was announced around one and a half year before its implementation. Therefore, we test for announcement effects on credit supply. Since syndicated loans are non-marketable assets, we do not expect to find a large announcement effect: until these assets can actually be posted into collateral accounts, they continue to occupy balance sheet space and can not be used as collateral.

Table B.2 presents results using the announcement instead of the implementation date. The dummy $Post05_t$ takes a value of zero until Q2 2005 and one thereafter. We find no significant effect across all specifications, which suggests that the announcement date did not play a relevant role in banks' lending decisions. Since syndicated bank loans are non-marketable, it is not surprising that there are no announcement effects of the collateral framework shock.

Table B.2: Announcement Effect. This table presents the effect of the single list's announcement date on credit supply. The analysis is based on data on the bank-loan-quarter level from Q1 2004 to Q4 2006. *Loan* is the loan issuance from bank i to firm j at quarter t. $Affected_i(0/1)$ is based on a median split of banks along the *Affected*-measure, see equation (2). $Post05_t$ indicates the announcement of the single-list regime after June 2005. The control variables on bank and loan level are lagged by 4 quarters. Country-fixed effects are based on respective headquarters. ***, **, * denote significance at the 1, 5, and 10% level, respectively.

VARIABLES	(1) ln(Loan Vol)	(2) ln(Loan Vol)	(3) ln(Loan Vol)	(4) ln(Loan Vol)
$\text{Affected}_i(0/1) \times \text{Post05}_t$	0.082	0.002	0.008	0.023
	(0.106)	(0.012)	(0.012)	(0.016)
Post05_t	0.253***			
	(0.093)			
Observations	12,163	12,163	12,163	12,029
R-squared	0.858	0.878	0.879	0.880
Bank-level Controls	No	No	Yes	Yes
Loan-level Controls	No	No	Yes	Yes
Bank × Firm FE	Yes	Yes	Yes	Yes
FirmCountry × Time FE	No	Yes	Yes	Yes
BankCountry × Time FE	No	Yes	Yes	Yes
FirmCountry × Industry × Time FE	No	No	No	Yes
Firm × Time FE	No	Yes	Yes	No
Cluster	Bank	Bank	Bank	Bank

B.3 Subsidiary Level

We also test for credit supply effects of the single list at the subsidiary level. We re-estimate (3) at the *subsidiary-level*:

$$ln(L_{kjt}) = \beta_1 \text{Affected}_k(0/1) \times \text{Post07}_t + \beta \text{X}_{i,t-4} + \mu_{kj} + \nu_{jt} + c_{jt} + \epsilon_{kjt} \ . \tag{8}$$

The median-split is now performed at the subsidiary level, such that a bank group i can have both affected and unaffected subsidiaries. Bank-firm fixed effects μ_{kj} are now defined at the subsidiary level, as is the clustering of standard errors. Since eligibility is restricted to loans involving at most two jurisdictions, we expect to observe significant effects at the subsidiary level as well.

Table B.3 shows the results of estimating the single list's effect on loan supply at the subsidiary level. Similar to the baseline specification, the coefficient of interest β_1 is significantly positive after including country $\times$ time and firm $\times$ time fixed effects and is robust to including loan-level controls and modifications in the fixed effect structure. Subsidiaries increase loan supply by 5.1% to 7.4%, depending on the specification: the effect size is slightly smaller than in the baseline but of comparable magnitude. The results are slightly weaker than using the *Affected*-measure at the group level, consistent with the existence of internal capital markets in multinational banks (De Haas and Van Lelyveld, 2010).

Table B.3: Credit Supply: Subsidiary Level. This table presents the effect of the single list's introduction on credit supply of subsidiaries. The analysis is based on data on the bank-loan-quarter level from Q1 2006 to Q4 2007. *Loan* is the loan issuance from subsidiary s to firm j at quarter t. *Affected$_k$(0/1)* is based on a median split of subsidiaries along the subsidiary-level *Affected*-measure, see equation (1). *Post07$_t$* indicates the single-list regime after January 2007. The loan-level control variables are lagged by 4 quarters. Country-fixed effects are based on respective headquarters. ***, **, * denote significance at the 1, 5, and 10% level, respectively.

Panel A: Subsidiary Measure

VARIABLES	(1) ln(Loan)	(2) ln(Loan)	(3) ln(Loan)	(4) ln(Loan)
Affected$_k$(0/1) × Post07$_t$	0.091	0.156	0.048*	0.046*
	(0.101)	(0.104)	(0.026)	(0.025)
Post07$_t$	0.146**			
	(0.067)			
Observations	9,904	9,904	9,904	9,800
R-squared	0.858	0.863	0.866	0.867
Loan-level Controls	No	No	Yes	Yes
Subsidiary FE	Yes	Yes	Yes	Yes
Subsidiary × Firm FE	Yes	Yes	Yes	Yes
FirmCountry × Time FE	No	Yes	Yes	Yes
SubCountry × Time FE	No	Yes	Yes	Yes
FirmCountry × Industry × Time FE	No	No	No	Yes
Firm × Time FE	No	Yes	Yes	No
Cluster	Subsidiary	Subsidiary	Subsidiary	Subsidiary

Panel B: Group Measure

VARIABLES	(1) ln(Loan Vol)	(2) ln(Loan Vol)	(3) ln(Loan Vol)	(4) ln(Loan Vol)
Affected$_i$(0/1) × Post07$_t$	0.124	0.061	0.077***	0.076***
	(0.104)	(0.148)	(0.029)	(0.029)
Post07$_t$	0.136**			
	(0.068)			
Observations	10,847	10,847	10,847	10,734
R-squared	0.859	0.863	0.866	0.867
Loan-level Controls	No	No	Yes	Yes
Subsidiary FE	Yes	Yes	Yes	Yes
Subsidiary × Firm FE	Yes	Yes	Yes	Yes
FirmCountry × Time FE	No	Yes	Yes	Yes
SubCountry × Time FE	No	Yes	Yes	Yes
FirmCountry × Industry × Time FE	No	No	No	Yes
Firm × Time FE	No	Yes	Yes	No
Cluster	Subsidiary	Subsidiary	Subsidiary	Subsidiary

B.4 Borrower Location - Triple Difference Specification

Table B.4: Borrower Location - Triple Diff. This table presents the effect of the single list's introduction on credit supply, interacted with a borrower location dummy. *Loan* is the loan issuance from bank i to firm j at quarter t. *Affected$_i$(0/1)* is based on a median split of banks along the *Affected*-measure, see equation (2). *Post07$_t$* indicates the single-list regime after January 2007. The control variables on bank and loan level are lagged by 4 quarters. Country-fixed effects are based on respective headquarters. ***, **, * denote significance at the 1, 5, and 10% level, respectively.

VARIABLES	(1) Lead ln(Loan Vol)	(2) Lead ln(Loan Vol)	(3) Participants ln(Loan Vol)	(4) Participants ln(Loan Vol)
Affected$_i$(0/1) × Post07$_t$ × Dom$_j$	0.275***	0.260***	-0.047**	-0.047**
	(0.041)	(0.066)	(0.022)	(0.022)
Affected$_i$(0/1) × Post07$_t$	0.007	0.005	0.013	0.013
	(0.023)	(0.023)	(0.023)	(0.023)
Dom$_j$ × Post07$_t$			0.053*	0.053*
			(0.029)	(0.029)
Observations	1,747	1,751	3,365	3,366
R-squared	0.882	0.883	0.892	0.892
Bank-level Controls	Yes	Yes	Yes	Yes
Loan-level Controls	Yes	Yes	Yes	Yes
Bank × Firm FE	Yes	Yes	Yes	Yes
FirmCountry × Time FE	Yes	Yes	Yes	Yes
BankCountry × Time FE	Yes	Yes	Yes	Yes
FirmCountry × Industry × Time FE	No	Yes	No	Yes
Firm × Time FE	Yes	No	Yes	No
Cluster	Bank	Bank	Bank	Bank

B.5 Alternative Affected-Measure

In this section, we present results using as an alternative measure affected defined relative to total bank assets rather than to loan supply. To do so, we use *total assets* at the group level

$$\text{AffectedTA}_i \equiv \frac{\sum_{t\in\mathcal{T}}\sum_{j\in\mathcal{J}^{ea}} L_{kjt}}{TA_{\text{Q2 2005}}} \cdot \text{Affected}_k \ . \tag{9}$$

This measure obtains from multiplying the baseline measure (2) by the group-level ratio of euro area loans to total assets. Banks to which the euro area loan market is of smaller importance are thus down-weighted under the modified measure (9). This alleviates concerns that results are driven by a control group of banks that are overall less active on the syndicated loan market but would lend almost exclusively to domestic borrowers if they became active.

Table B.5: Affected Defined Over Total Assets: Loan Supply and Spreads. This table presents the effect of the single list's introduction on credit supply and loan spreads. The analysis is based on data on the bank-loan-quarter level from Q1 2006 to Q4 2007. *Loan* is the loan issuance from bank i to firm j at quarter t. *AffectedTA$_i$(0/1)* is based on a median split of banks along the modified *Affected*-measure, see equation (9). *Post07$_t$* indicates the single-list regime after January 2007. The control variables on bank and loan level are lagged by 4 quarters. Country-fixed effects are based on respective headquarters. ***, **, * denote significance at the 1, 5, and 10% level, respectively.

VARIABLES	(1) ln(Loan Vol)	(2) ln(Loan Vol)	(3) Spread	(4) Spread
Affected (0/1) × Post07	0.032**	0.017	-5.087**	-5.863**
	(0.015)	(0.019)	(2.285)	(2.338)
Observations	10,085	10,043	10,085	10,043
R-squared	0.862	0.864	0.760	0.761
Bank-level Controls	Yes	Yes	Yes	Yes
Loan-level Controls	Yes	Yes	Yes	Yes
Bank × Firm FE	Yes	Yes	Yes	Yes
FirmCountry × Time FE	Yes	Yes	Yes	Yes
BankCountry × Time FE	Yes	Yes	Yes	Yes
FirmCountry × Industry × Time FE	No	Yes	No	Yes
Firm × Time FE	Yes	No	Yes	No
Cluster	Bank	Bank	Bank	Bank

B.6 Excluding German Banks

Since German banks have been comparatively active on the international loan market, particularly in the US, a large share of affected banks might be located in Germany, which could confound our identification strategy. To ensure that our results are not driven by German banks, we re-classify all non-German banks into *affected* and *unaffected* banks and re-estimate equation (3). Table B.6 shows that our baseline loan supply results is robust to excluding German banks.

Table B.6: Excluding German Banks: Loan Supply. This table presents the effect of the single list's introduction on credit supply. The analysis is based on data on the bank-loan-quarter level from Q1 2006 to Q4 2007. *Loan* is the loan issuance from bank i to firm j at quarter t. $AffectedTA_i(0/1)$ is based on a median split of banks along the modified *Affected*-measure, see equation (9). $Post07_t$ indicates the single-list regime after January 2007. The control variables on bank and loan level are lagged by 4 quarters. Country-fixed effects are based on respective headquarters. ***, **, * denote significance at the 1, 5, and 10% level, respectively.

VARIABLES	(1) ln(Loan Vol)	(2) ln(Loan Vol)	(3) ln(Loan Vol)	(4) ln(Loan Vol)
Affected w/o $DE_i(0/1) \times Post07_t$	0.066	0.041**	0.066*	0.079**
	(0.063)	(0.020)	(0.040)	(0.035)
$Post07_t$	0.202***			
	(0.048)			
Observations	7,304	7,304	7,304	7,208
R-squared	0.852	0.863	0.864	0.865
Bank-level Controls	No	No	Yes	Yes
Loan-level Controls	No	No	Yes	Yes
Bank × Firm FE	Yes	Yes	Yes	Yes
FirmCountry × Time FE	No	Yes	Yes	Yes
BankCountry × Time FE	No	Yes	Yes	Yes
FirmCountry × Industry × Time FE	No	No	No	Yes
Firm × Time FE	No	Yes	Yes	No
Cluster	Bank	Bank	Bank	Bank